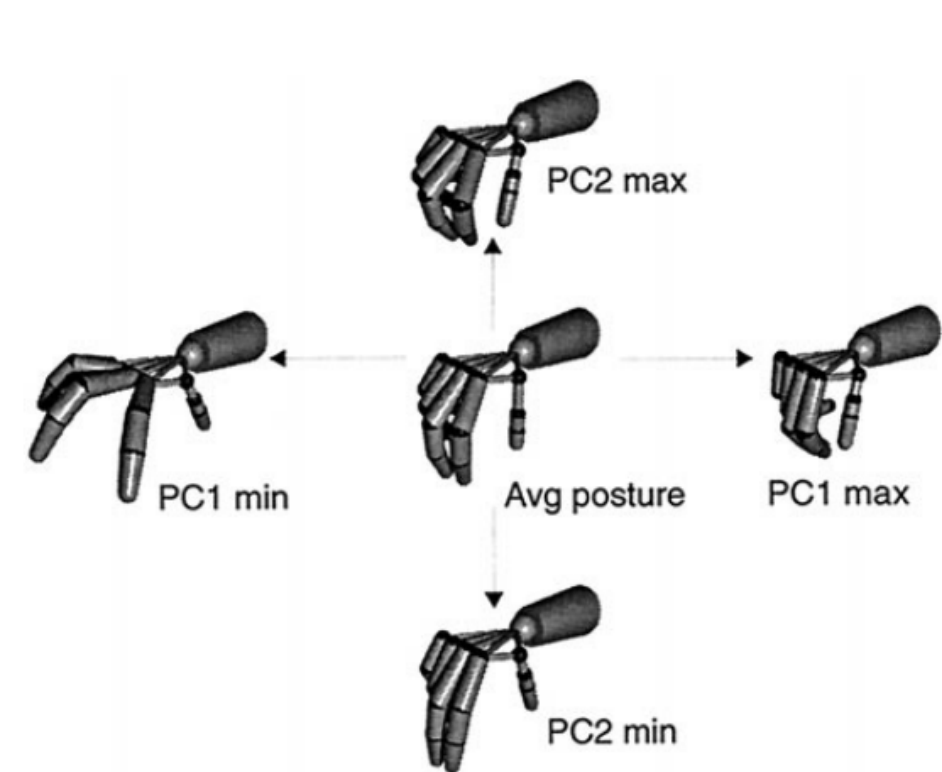


Modeling Adaptive Grasping in a String-Driven Origami Hand

Introducing FMAS: a friction-modulated tendon transmission model connected to MuJoCo contact simulation
Yushuo Wang

Problem Statement

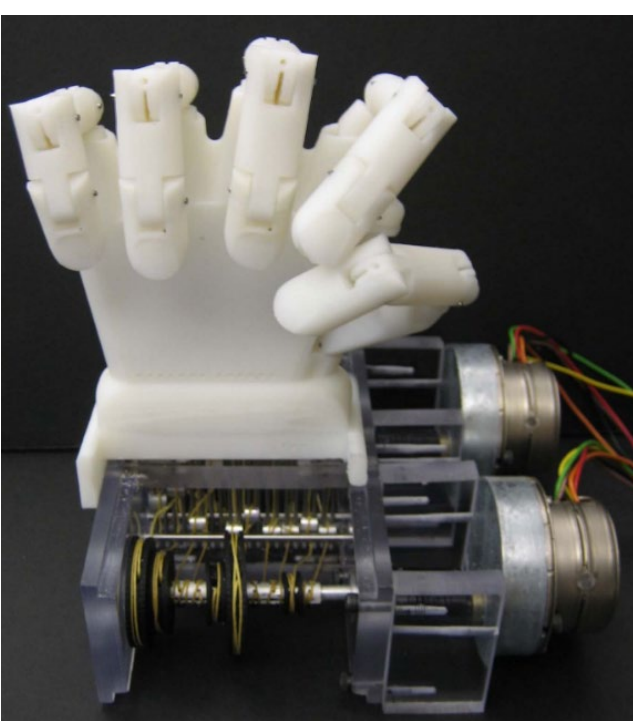


the first 2 PCs of human hand grasping

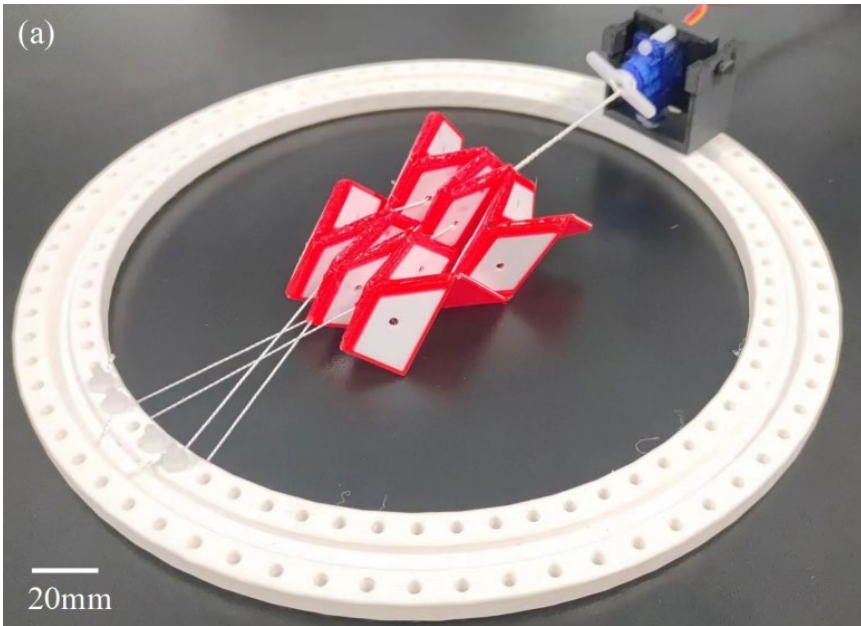
Postural Synergy Theory

Human hand motion has many degrees of freedom, but grasping postures are not controlled as independent joint commands. A classical observation in hand neuroscience is that many hand shapes can be approximated by a small number of postural synergies: low-dimensional variables coordinate high-dimensional joint configurations.

The origami hand is not treated as a conventional fully actuated kinematic linkage. Its folds, compliant creases, and routed tendons create a differential-like underactuated mechanism. In a two-ended closed-loop tendon, pulling from side A and pulling from side B are not equivalent because tension propagates through different routed paths. A fixed transmission matrix can describe one nominal reference motion, but it cannot explain this direction-dependent transmission. Therefore, this project introduces a two-ended tendon-driven synergy model and connects it to numerical contact simulation.



mechanical implementation of synergy



String-driven origami

The simulation problem has two connected levels. The geometric and kinematic level explains what motion the origami hand design and tendon routing are intended to produce. The physical simulation level then asks how this reference motion is realized when the hand is represented as an articulated model and advanced through numerical integration and contact constraints. MuJoCo is used to solve the time response $q(t)$, contact count, contact force, and object interaction.

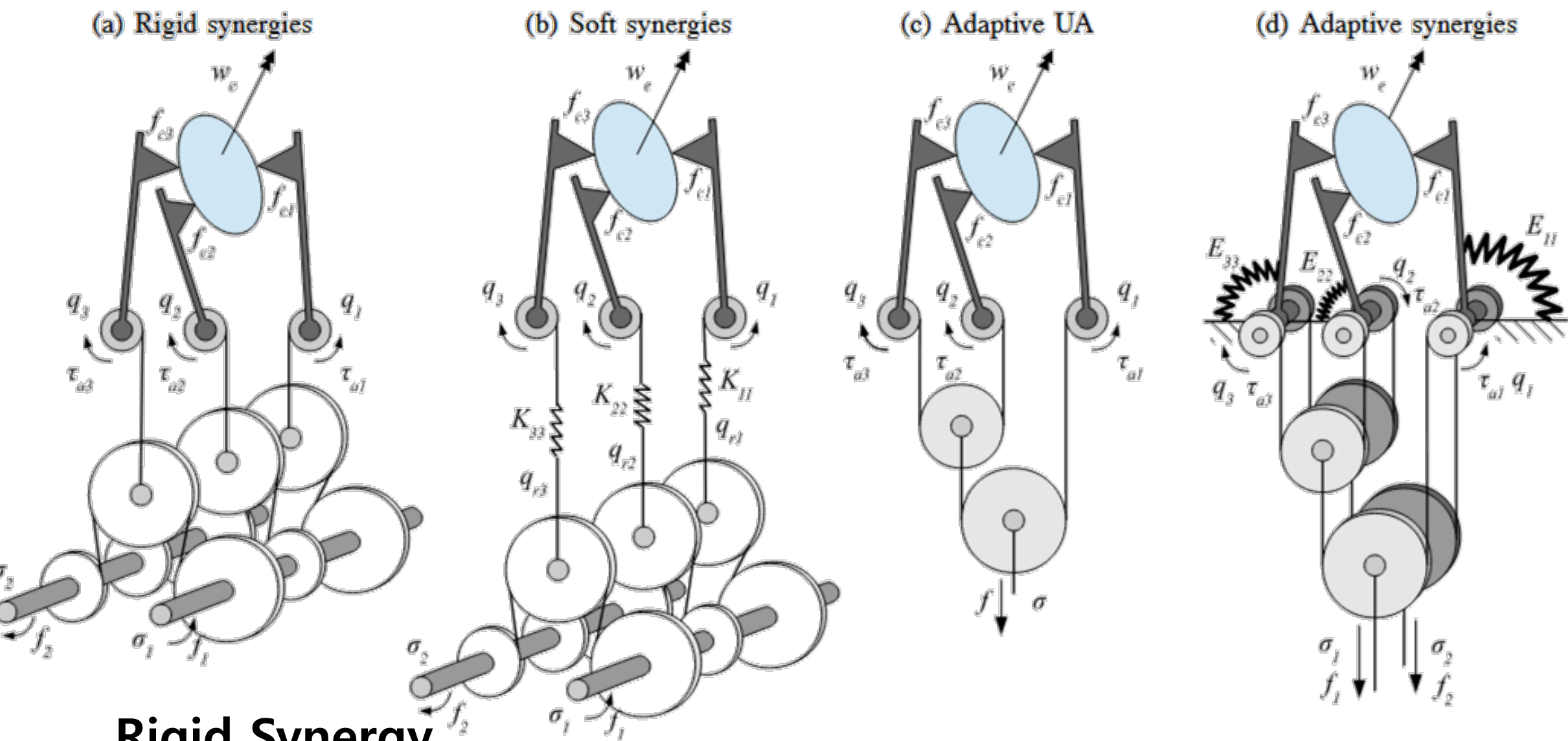
The central simulation question is: **how can a string-driven origami hand design be simulated across geometry, synergy transmission, numerical dynamics, and contact.**

Literature Review

Postural synergy studies show that high-dimensional hand postures can often be represented by a few low-dimensional coordinates. This motivates **synergy-driven hands**: a small number of commands can organize many joints into coordinated grasping motions.

Soft synergy models add an important physical interpretation. A synergy command should not be treated as a rigid final posture, but as a compliant reference. When the hand contacts an object, the realized posture can deviate from the reference due to external forces and structural compliance.

Adaptive synergy demonstrate how this idea can be implemented mechanically. A tendon transmission maps actuator displacement into multiple joint coordinates, while passive compliance allows the hand to adapt during contact.



Rigid Synergy

$$q = S^{(k)} \sigma^{(k)}$$

q : joint configuration
 $S^{(k)}$: synergy matrix

Core idea:

A high-dimensional hand posture can be approximated by a low-dimensional set of fixed synergy coordinates.

Soft Synergy

$$q_r = S^{(k)} \sigma^{(k)}$$
$$J^T f_c = K(q_r - q)$$

q_r : joint configuration of the reference posture
 J : Jacobian of the hand
 f_c : contact force

Core idea:

The synergy command defines a reference posture, while compliance and contact forces allow the actual hand shape to deviate from it.

Adaptive Synergy

$$Rq = x$$
$$q = CJ^T f_c + S^{(k)} \sigma^{(k)}$$

R : transmission matrix of underactuation mechanism

Core idea:

A mechanical transmission maps a small number of actuator displacements into many joint coordinates, realizing synergy through underactuation.

Methodology

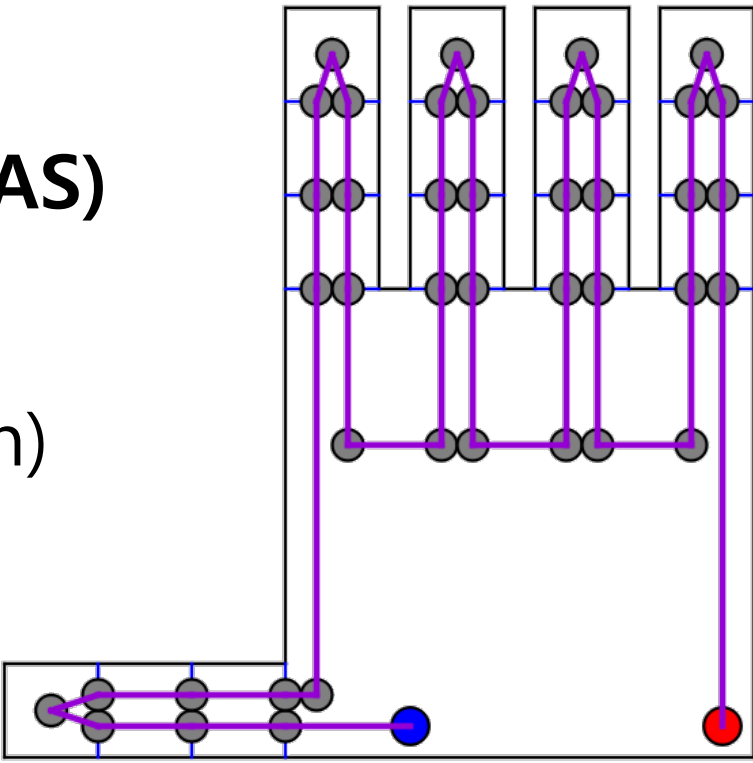
Friction-Modulated Adaptive Synergy(FMAS)

$$T_j^{(A)} = \alpha_j^{(A)} F_A$$

$$\alpha_j^{(A)} = \exp\left(-\sum_{k \in P_A(j)} \beta_k\right) \quad (\text{Capstan friction})$$

$$J^T f_c = R_A^T F_A + R_B^T F_B - E q$$

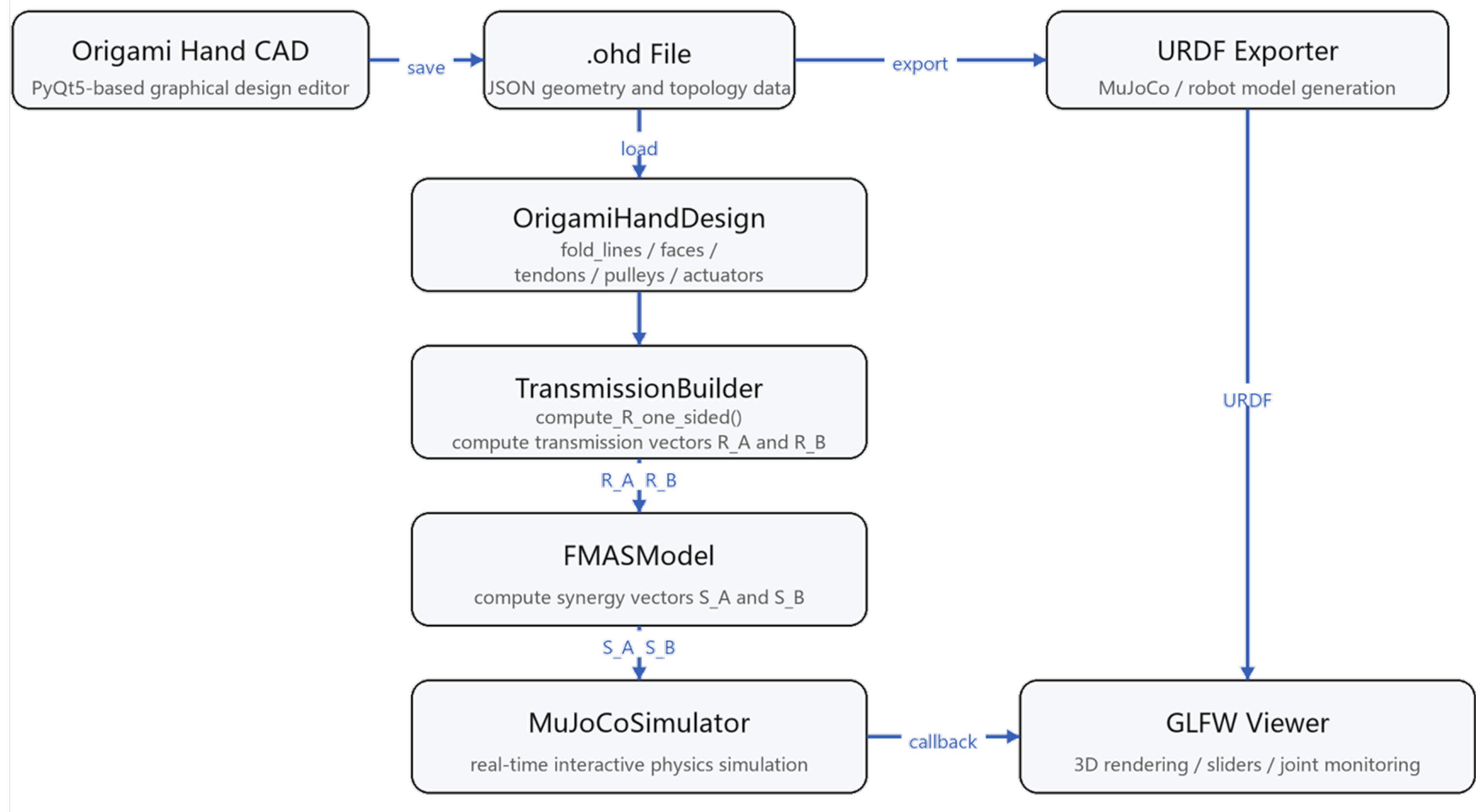
$$q = S_A x_A + S_B x_B + C J^T f_c$$



a two-ended closed-loop tendon driven hand

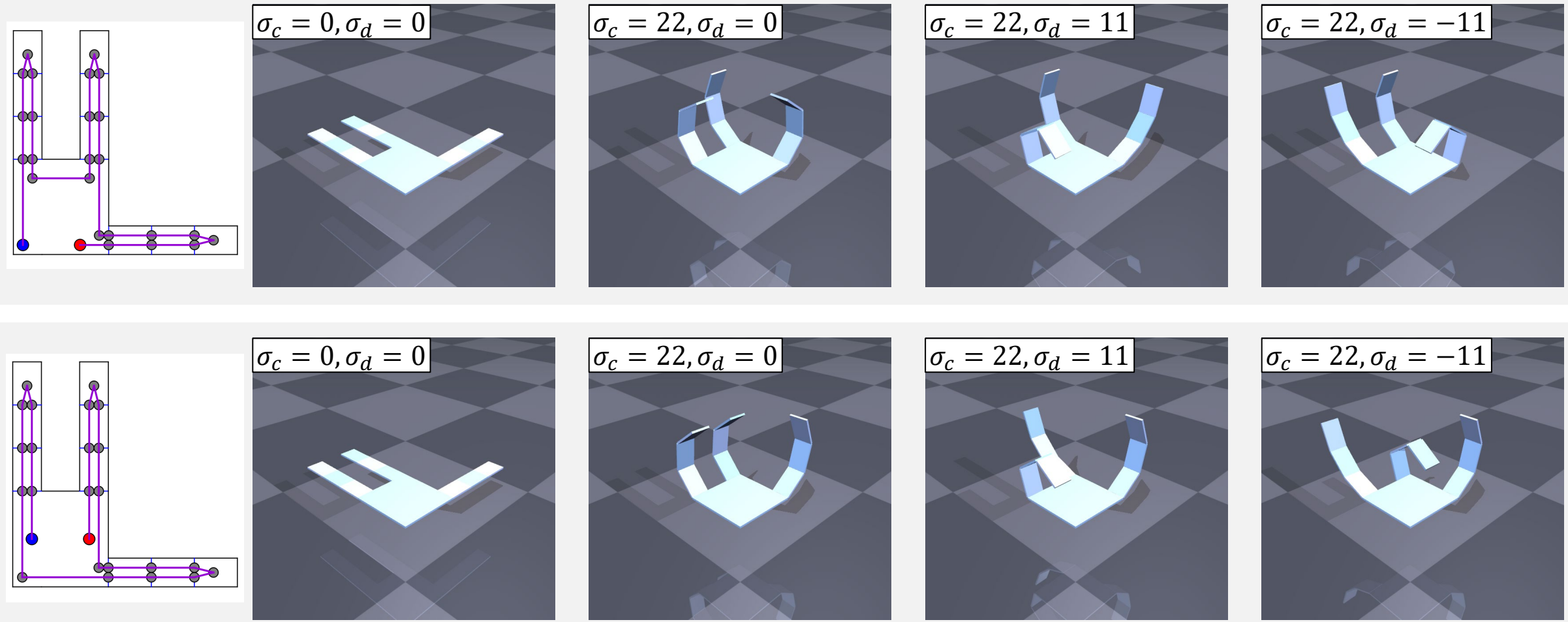
The **.ohd** design file defines the origami hand geometry: faces, creases, joints, holes, actuators, and routed tendons.

The FMAS layer converts the two-ended tendon routing into a distribution matrix. Pulling from side A and side B gives two one-sided transmissions, R_A and R_B , which generate two synergy bases.



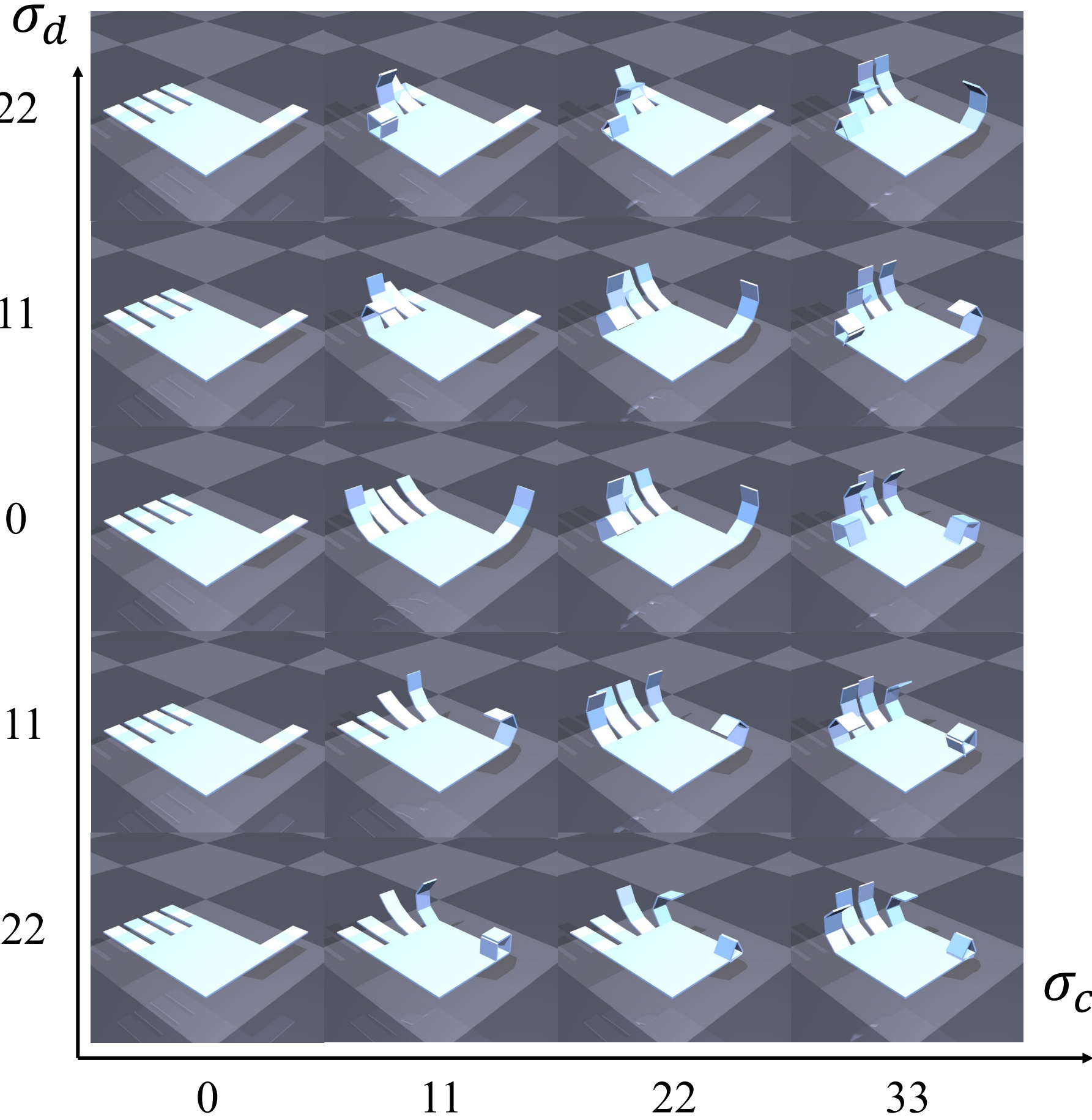
Experimental Results

Case I



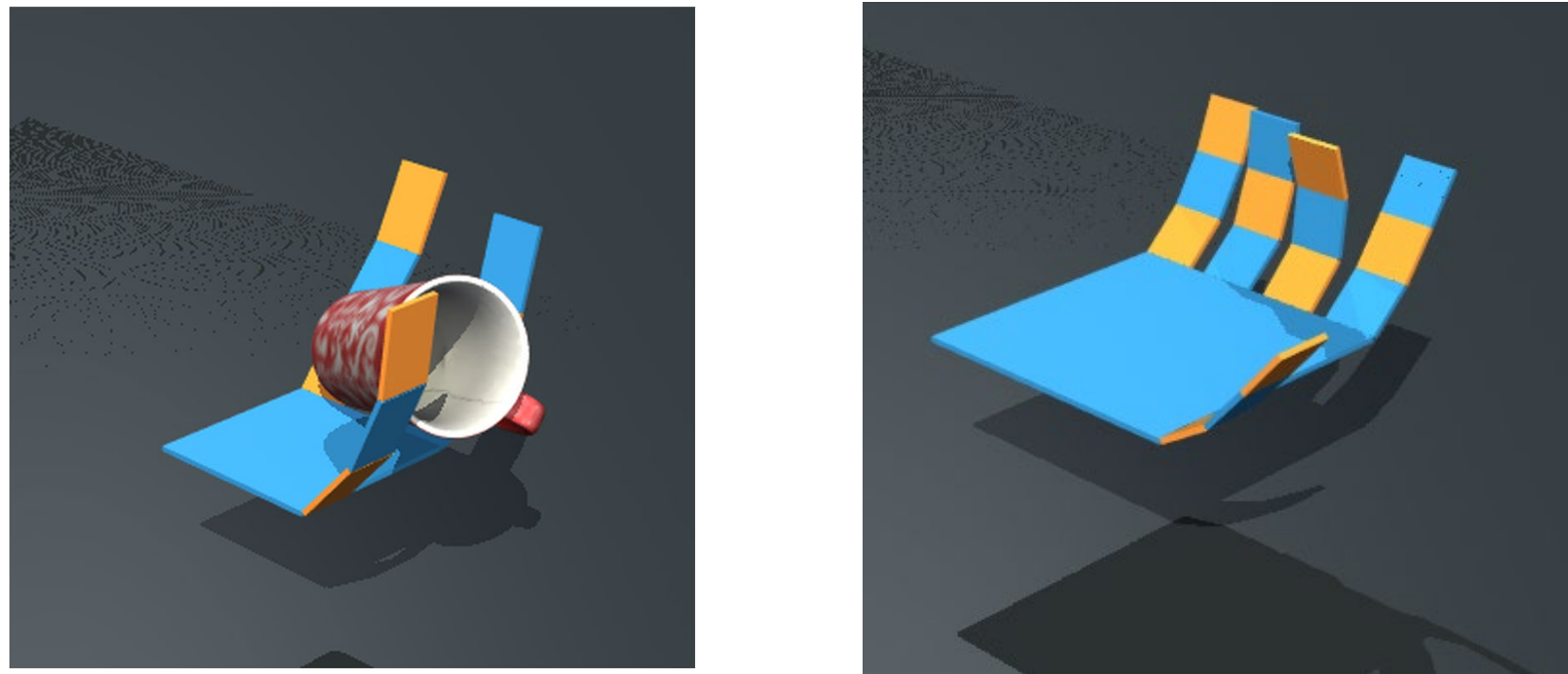
simulation results for a three-finger gripper with two different tendon routes

Case II



simulation results for a five-finger origami hand

Case III



grasping contact numerical simulation results

Conclusion

This project presents a physics-aware simulation framework for a string-driven origami dexterous hand. The geometric layer converts .ohd designs into hand topology, URDF/STL geometry, tendon routing, and reference synergy motion. The proposed FMAS model derives physically interpretable closing and redistribution modes from the two-ended closed-loop tendon transmission, instead of prescribing a fixed synergy basis. The MuJoCo layer converts these model-derived inputs into numerical time integration and contact simulation, producing joint trajectories, contact statistics, screenshots, GIFs, and auditable MJCF files.